The Crucial Rule: `this` Behaves Differently

Inside a standard instance method, the keyword `this` refers to the specific instance object (`bike1`).

However, inside a `static` method, `this` refers directly to the class constructor itself (`BikeFactory`). Because of this, a static method cannot access instance data like `this.model`, since no specific instance is active when the static method runs.

Context	What `this` refers to
Instance method	The specific instance (`bike1`)
Static method	The class constructor itself (`BikeFactory`)

When Should You Use `static`?

You should reach for `static` whenever you need:

1. **Utility Helpers**

Functions that perform calculations but don't need to read or alter any unique object state (e.g., `Math.max()` or `Date.now()` are built-in static methods).

2. **Global Trackers / Factories**

Keeping records of configurations, counts, or handling caching limits that span across all instances collectively.

> Rule of thumb: if the method needs `this.someProperty` from a specific object, keep it an instance method. If it's about the class as a whole — not any one object — make it static.